

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA16 COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3, BL4, BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 30

Total Marks:30

Annexure A

MCQ TEST

Criteria	Conceptual Understanding (3)	Accuracy of Answers (2.5)	Application of Knowledge (2)	Time Management (1.5)	Question Interpretation (1)	Total(10)
<i>Weightage</i>	30%	25%	20%	15%	10%	100%
<i>Q</i>						

Code of Conduct:

I ____Ajay I /192421147____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Ajay I

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding ; several incorrect responses	Lacks understanding ; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ANNEXURE A

MCQ Test:

1.Consider a simple linear regression model with one independent variable. Which of the following is true? (CO3)

Option_a: The regression line always passes

2.Which of the following statements is correct about Decision Trees? (CO3)

Option_c: Decision trees can be used for classification and regression.

3.Which of the following methods can be used for pruning in decision trees? (CO3)

Option_c: Both (a) and (b)

4.Which of the following is a disadvantage of Decision Trees? (CO3)

Option_a: Prone to overfitting.

5.What does Bayes' Theorem primarily help with in probability and statistics? (CO3)

Option_b: Updating probabilities based on new evidence

6.In Bayes' Theorem, what does the term $P(A|B)$ represent? (CO3)

Option_b: The probability of event A given event B has occurred

7.A factory has two machines, A and B. Machine A produces 60% of the products, and machine B produces 40%. If 2% of A's products and 3% of B's products are defective, what is the probability that a randomly selected defective product came from Machine B? (CO3)

Option_c: 0.60

8.Which of the following is not required to apply Bayes' Theorem? (CO3)

Option_c: Total number of possible outcomes

9.A doctor uses a test to diagnose a disease that affects 1% of a population. The test correctly identifies the disease 95% of the time but also gives a false positive in 5% of cases. If a patient tests positive, what is the probability they actually have the disease? (CO3)

Option_a: 0.16

10.Bayes' Theorem is most useful when dealing with which type of probability? (CO3)

Option_b: Conditional probability

11.If a hypothesis HHH has a prior probability of 0.3 and new evidence EEE has a likelihood of 0.8 given HHH, how does Bayes' Theorem adjust the probability of HHH after seeing EEE? (CO3)

Option_a: The probability of HHH increases

12.A bag contains 5 red and 7 blue balls. A ball is drawn randomly and found to be red. What is the probability that the next ball drawn is also red (assuming the first ball is not replaced)? (CO3)

Option_b: 4/11

13.Which field heavily relies on Bayes' Theorem for updating beliefs based on new data? (CO3)

Option_a: Machine Learning

14.Why is backpropagation crucial in training deep neural networks? (CO3)

Option_b: It updates weights using the gradient of the loss function.

15.During backpropagation, what happens when the learning rate is too high? (CO3)

Option_b: The model may overshoot the optimal weight values and fail to converge.

16.What is the main advantage of rule-based induction models? (CO3)

Option_a: They provide an interpretable set of rules for decision-making.

17.Which of the following is an example of a rule-based induction algorithm? (CO3)

Option_a: Decision Tree

18.In rule-based induction, what happens when the rules are too specific to the training data?

Option_a: The model may suffer from overfitting.

19. Which approach is commonly used to simplify rule-based induction models? (CO3)

Option_a: Pruning redundant or less significant rules

20. What is the primary purpose of a confusion matrix in evaluating a classifier? (CO3)

Option_a: To visualize the performance of a classifier by showing correct and incorrect predictions

21. Which metric is used to measure the proportion of correctly classified instances out of the total instances? (CO3)

Option_c: Accuracy

22. In a binary classification problem, what does the True Positive (TP) value represent? (CO3)

Option_b: The number of positive instances correctly classified as positive

23. What is the formula for calculating precision in a classification problem? (CO3)

Option_a: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

24. Which of the following metrics is most useful when the classes are imbalanced? (CO3)

Option_b: F1-Score

25. What does a high recall value indicate in a classification model? (CO3)

Option_b: The model has a low number of false negatives

26. Which of the following is an example of a rule-based induction algorithm? (CO3)

Option_a: Decision Tree

27. In rule-based induction, what happens when the rules are too specific to the training data? (CO3)

Option_a: The model may suffer from overfitting.

28. Which approach is commonly used to simplify rule-based induction models? (CO3)

Option_a: Pruning redundant or less significant rules

29. What is the primary purpose of a confusion matrix in evaluating a classifier? (CO3)

Option_a: To visualize the performance of a classifier by showing correct and incorrect predictions

30. Which metric is used to measure the proportion of correctly classified instances out of the total instances? (CO3)

Option_c: Accuracy